

The Twelfth Thought Experiment: From the Fine-Structure Constant to Observation Maps and Gauge Symmetry

— Starting from the Fine-Structure Constant, Rereading Spacetime, Mass, Charge, Color Charge and Gauge Symmetry from the Light-Cone Equation and the Choice of Observable Axes —

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Related papers: Design document (Paper 0) [S1], the Eleventh Thought Experiment [S2], nontrivial quadratic closure [S3], reexamination of xyztRQ [S4], composition of central projection [S5]

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Abstract

In the previous paper, the Eleventh Thought Experiment [S2], the fine-structure constant α was used as a known input value as it stands. This paper starts by asking where that α comes from. In the Bohr ground state of hydrogen $\alpha = v/c$, so α can first be read as a velocity ratio. The Lorentz factor $\gamma = 1/\sqrt{1 - \alpha^2}$ is only a component ratio of $r^2 + \tau^2 + (it)^2 = 0$, which is the light-cone equation $x^2 + y^2 + z^2 + (it)^2 = 0$ with one proper-time axis added. A worldline slower than light lies on the light cone in the space with one more axis.

We extend this light-cone equation to eight directions, the three spatial directions plus the not directly readable R, Q_1, Q_2, Q_3 : $x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2 + (it)^2 = 0$. We give the axes no physical names from the start and treat them as eight unnamed directions $(u_1, \dots, u_8)$. When an observer chooses a directly readable subspace, the remaining directions cannot be distinguished and appear only as a sum of squares. A transformation G for which the observation map P satisfies $P(U) = P(GU)$ is an observational symmetry, so symmetry can be read as the mathematical expression of indistinguishability. An observer who reads only t sees an $O(7)$ -type symmetry, and if the five hidden directions split as 3+2, $S(U(3) \times U(2)) \simeq (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ appears as a candidate. This is the same structure as the 3+2 split of the SU(5) grand unified theory.

This paper is a conceptual note without computation; the eight dimensions, the signature and the 3+2 split from §4 on are put in by hand as assumptions.

0. Position of this paper

The Eleventh Thought Experiment [S2] computed two-body systems of standard theory numerically, referenced only to the rest frame of an absorber at infinity. The input α was the CODATA value [1], and its meaning was not asked.

This paper is the work in the opposite direction. It does not accept α as an input value; it starts from “what is α in the first place”, returns to the light-cone equation, removes the names of the axes, and rereads spacetime, mass, charge, color charge and gauge symmetry from what an observer can distinguish. It is also a rewriting, in geometric language, of the first-principle candidate of the design document [S1], “states exist that have no names but are distinguishable from one another and can be counted”, and of the position that “a state itself cannot be read directly”.

This paper is a conceptual note without computation; the eight dimensions, the signature and the 3+2 split from §4 on are put in by hand as assumptions.

1. Introduction

This thought experiment started from a very simple question.

Where does the fine-structure constant α come from in the first place?

In the standard treatment it is defined as

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137.036}$$

[1], and in quantum electrodynamics it is treated as the dimensionless coupling constant that expresses the strength of the electromagnetic interaction.

But if we go back to the historical starting point [2, 3, 4], in the Bohr model of the hydrogen atom the ground state satisfies

$$\frac{v}{c} = \alpha.$$

That is,

$\alpha = \frac{\text{velocity of the electron in the hydrogen ground state}}{\text{speed of light}}$

Therefore

$$\alpha^{-1} \simeq 137$$

can first be read, naively, as the velocity ratio

$$c \simeq 137 v.$$

Starting from this simple fact, might we be able to reread the fine-structure constant, special relativity, the Minkowski metric, and even mass, charge, color charge and gauge symmetry from a different angle?

This paper follows that question step by step.

2. Reading the fine-structure constant as a velocity ratio

In the Bohr model of the hydrogen atom [2],

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

and

$$mvr = \hbar$$

give

$$v = \frac{e^2}{4\pi\epsilon_0 \hbar}.$$

Dividing by the speed of light c ,

$$\frac{v}{c} = \frac{e^2}{4\pi\epsilon_0 \hbar c} = \alpha.$$

Therefore in the hydrogen ground state

$$\boxed{\frac{v}{c} = \alpha}$$

It is then no surprise that α appeared when Sommerfeld put special-relativistic corrections into the hydrogen atom [3].

The Lorentz factor of special relativity is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

so substituting

$$\frac{v}{c} = \alpha$$

gives

$$\gamma = \frac{1}{\sqrt{1 - \alpha^2}}.$$

In the low-velocity approximation,

$$\gamma = 1 + \frac{1}{2}\alpha^2 + \frac{3}{8}\alpha^4 + \dots$$

In this form, the reason why $\alpha^2, \alpha^4, \dots$ appear in the fine structure is very intuitive.

But here an even simpler view is possible.

Why do we have to think from γ at all?

3. Seeing it from the light-cone equation rather than the Lorentz factor

Set $c = 1$. The propagation of light is expressed by Minkowski's light-cone equation

$$x^2 + y^2 + z^2 - t^2 = 0.$$

Writing time as the imaginary number it [5],

$$x^2 + y^2 + z^2 + (it)^2 = 0$$

The Lorentz transformations are determined as the linear transformations that preserve this form. What comes first is the light-cone equation; the Lorentz transformations appear afterwards as the transformations that preserve it.

Next consider a particle moving with velocity v . With the distance traveled r and the proper time τ ,

$$t^2 - r^2 = \tau^2.$$

Moving τ^2 to the left-hand side gives

$$r^2 + \tau^2 + (it)^2 = 0$$

This has the same form as the light-cone equation. A worldline that is slower than light in (r, t) lies on the light cone in (r, τ, t) , where one proper-time axis τ has been added [6, 11] (Fig. 1(a)(b)).

Dividing by t^2 ,

$$\left(\frac{r}{t}\right)^2 + \left(\frac{\tau}{t}\right)^2 = 1.$$

Reading here

$$\frac{r}{t} = \frac{v}{c} = \alpha,$$

we have

$$\left(\frac{\tau}{t}\right)^2 + \alpha^2 = 1$$

and

$$\frac{\tau}{t} = \sqrt{1 - \alpha^2}.$$

Taking the reciprocal,

$$\gamma = \frac{t}{\tau} = \frac{1}{\sqrt{1 - \alpha^2}}.$$

Therefore γ is not fundamental.

What comes first is

$$\boxed{r^2 + \tau^2 + (it)^2 = 0},$$

the light-cone equation with one axis added, and the Lorentz factor is nothing but its component ratio read from one axis.

In this paper we call a zero sum of squares of this form, containing one imaginary component, a **quadratic closure**. The name belongs to this paper, but the equation is the light-cone equation itself. That a zero sum of squares can have a nontrivial solution only if at least one component is not real was shown in [S3].

Here the way the problem looks changes.

4. Extending spacetime

Next, using ordinary three-dimensional space and splitting the τ^2 of §3 into two directions R and Q , we put

$$\boxed{x^2 + y^2 + z^2 + R^2 + Q^2 = t^2}$$

Moving terms,

$$x^2 + y^2 + z^2 + R^2 + Q^2 + (it)^2 = 0.$$

This is also the light-cone equation, only with more axes. The imaginary axis is t alone.

What matters here is not to fix the physical meaning of each axis from the start.

First, simply, we read

$$(x, y, z)$$

as the directions that can be measured directly as space, and

$$(R, Q)$$

as the directions that cannot be measured directly in the same way.

Then

$$t^2 - (x^2 + y^2 + z^2) = R^2 + Q^2$$

can be read as the observational relation

$$\begin{aligned} &\text{difference of squares of time and the directly readable part} \\ &= \text{sum of squares of the not directly readable part} \end{aligned}$$

For an observer who reads xyz , this equation has the same form as the equation of central projection. When a point $P = (x, y, z)$ on the tangent plane at distance R_c from the center O is mapped to the sphere of radius R_c by $\pi(P) = R_c P/\ell$ [S5], the distance ℓ from O to P satisfies

$$\ell^2 = x^2 + y^2 + z^2 + R_c^2.$$

Comparing with the equation above, the curvature radius in the projection direction is the square root of the sum of squares of the not directly readable directions, $R_c = \sqrt{R^2 + Q^2}$, and the distance ℓ from the center to the point corresponds to t . The proper time τ of §3 corresponds to R_c , the spatial distance r to the distance $|x|$ on the tangent plane, and the slope $v = r/t$ of the worldline coincides with $\sin \theta = |x|/\ell$ of the angle θ of the projection ray. Removing the imaginary unit i from the light-cone equation gives the Pythagorean equation of central projection (Fig. 1(c)).

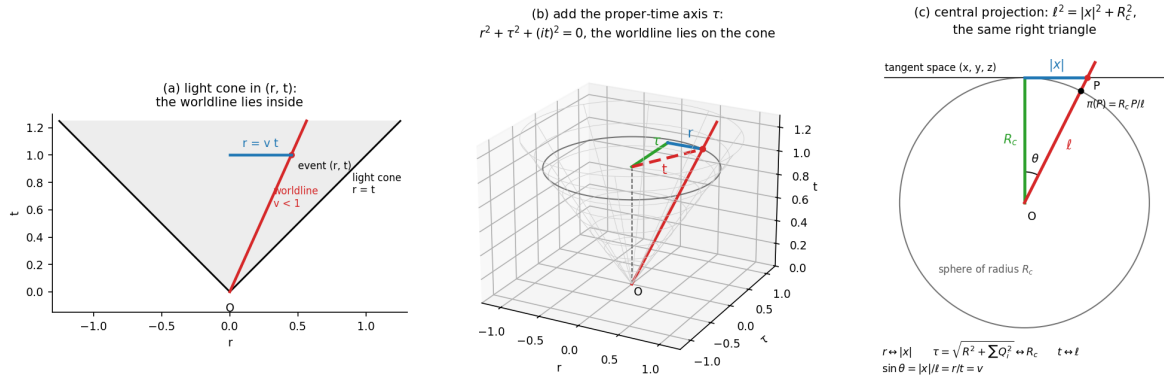


Figure 1. The light cone and central projection are the same right triangle. (a) The light cone in the (r, t) plane. A worldline of slope $v < 1$ lies inside the cone. (b) Adding the proper-time axis τ gives $r^2 + \tau^2 + (it)^2 = 0$, and the same worldline lies on the surface of the cone. In the cut at time t , a right triangle forms with legs r (blue) and τ (green) and the cut radius t (red) as hypotenuse. (c) Central projection $\ell^2 = |x|^2 + R_c^2$. The same triangle appears as the distance $|x|$ on the tangent plane (blue), the curvature radius R_c in the projection direction (green), and the distance ℓ from the center to the point (red). $r \leftrightarrow |x|$, $\tau = \sqrt{R^2 + \sum Q_i^2} \leftrightarrow R_c$, $t \leftrightarrow \ell$, $\sin \theta = v$. A schematic drawn for a generic v ; there are no data.

At this point we do not need to assume that

- t is essentially a “time axis”
- R is essentially a “mass axis”
- Q is essentially a “charge axis”.

These may be names attached afterwards to observational results. The names t, R, Q are those used in [S4]. The reading that relates the fifth coordinate to rest mass also appears in Wesson’s Space-Time-Matter theory [12].

5. Perhaps there is only one kind of momentum

Suppose a state is expressed by one phase

$$\Phi.$$

Then the momentum in each direction can in general be written as its phase gradient,

$$K_A = \partial_A \Phi.$$

For the ordinary spatial directions we read it as

$$(p_x, p_y, p_z).$$

Applying the same idea to the other axes, there is only one gradient vector

$$K = (K_x, K_y, K_z, K_t, K_R, K_Q).$$

It is possible that what we call

$$K_t \rightarrow E,$$

$$K_R \rightarrow m,$$

$$K_Q \rightarrow q$$

are its components.

Applying the closure of §4 to this gradient gives

$$K_t^2 = K_x^2 + K_y^2 + K_z^2 + K_R^2 + K_Q^2,$$

that is,

$$E^2 = p^2 + m^2 + q^2,$$

which returns to the ordinary mass shell $E^2 = p^2 + m^2$ for $q = 0$. The reading in which the momentum in a hidden direction is read as charge and its square is added to the mass shell is the same structure as in Kaluza-Klein theory [7, 8], where a particle moving on the light cone in five dimensions carries its momentum in the hidden direction as charge in four dimensions and its square as the four-dimensional mass.

But here too the order has to be reversed.

What really exists from the start is not

$$E, m, q.$$

There are first unnamed gradient components, and we may merely be giving the names

energy, mass, charge

to the ways they appear in observation.

Then the view

momentum, energy, mass and charge are
projections of one gradient onto different directions

becomes possible.

6. Splitting Q into three directions

There is no need to take Q as a single axis either.

Rather, we put the three directions

$$(Q_1, Q_2, Q_3).$$

The whole is then the eight-dimensional

$$(x, y, z, t, R, Q_1, Q_2, Q_3)$$

The quadratic closure is

$$x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2 = t^2$$

or

$$x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2 + (it)^2 = 0$$

It is the light-cone equation in eight dimensions. For an observer who reads xyz , the curvature radius of the central projection is $R_c = \sqrt{R^2 + Q_1^2 + Q_2^2 + Q_3^2}$.

It has been shown in the Kaluza-Klein framework that obtaining the Standard Model group as isometries of the hidden directions requires at least seven directions outside the four spacetime directions [13]. In this paper the directions outside the four spacetime directions are the four R, Q_1, Q_2, Q_3 , which is not enough as isometries. This paper reads the group not from isometries but from indistinguishability (§9-§11). A precedent that reads the Standard Model group from eight dimensions (the octonions) is Furey [14].

Here too we do not think that the physical meanings

$$x, y, z, t, R, Q_1, Q_2, Q_3$$

exist from the start.

Originally we think there are only

$$(u_1, u_2, \dots, u_8),$$

eight unnamed directions.

When an observer can read three of them directly as space, it names them

$$(u_1, u_2, u_3) \rightarrow (x, y, z).$$

It may merely be naming the features observed in the remaining directions

$$t, \quad R, \quad Q_1, Q_2, Q_3.$$

Here the meaning of physical quantities itself becomes an observer-dependent map.

7. Observers who can read different axes

From here a strong thought experiment becomes possible.

That we directly read

$$(x, y, z)$$

does not mean that this is the only possible observation map.

For example, an observer that can directly read only

$$(Q_1, Q_2, Q_3)$$

can be defined mathematically.

For that observer,

$$(Q_1, Q_2, Q_3)$$

looks like ordinary three-dimensional space.

Conversely,

$$x, y, z, t, R$$

become internal directions that cannot be read directly.

The physical phenomena of that world would look extremely different from those of ours.

But if the state and the interactions on the eight-dimensional side are the same, this is nothing but

different observation maps of the same physics

This is not “different physical laws” but

the result of choosing a different subspace as the observed space

8. What is seen if only the t axis can be read

Consider an even more extreme case.

Consider an observer that can directly read, of the eight directions, only

$$t.$$

The remaining seven directions,

$$x, y, z, R, Q_1, Q_2, Q_3,$$

cannot be distinguished individually.

What can then be observed is not their breakdown but only the sum of squares

$$\rho^2 = x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2$$

That is, it appears only as

$$t^2 = \rho^2$$

To this observer, the eight-dimensional light cone is seen only as its radius $\rho = t$.

A complicated state with seven degrees of freedom degenerates into one scalar quantity.

This is important.

The unobservable degrees of freedom have not disappeared.

Because they cannot be distinguished, they appear only as one norm.

In general, if only k of the eight dimensions can be read directly,

$$\mathbb{R}^8 \rightarrow \mathbb{R}_{\text{observed}}^k \oplus \mathbb{R}_{\text{hidden}}^{8-k}.$$

If the hidden side cannot be distinguished individually, it is observed as one quantity

$$\rho_{\text{hidden}}^2 = \sum_{\text{hidden}} u_a^2.$$

Therefore

dimensional reduction is not that dimensions
physically disappear, but that the observer
can no longer distinguish them

can be read.

9. Symmetry appears from here

This thought experiment gives another view of why standard theory requires strong symmetries.

Let the observation map be

$$P.$$

If, after acting with a transformation G on a state U ,

$$P(U) = P(GU),$$

then the observer cannot distinguish the two.

Therefore

$$P(U) = P(GU) \implies G \text{ is an observational symmetry transformation}$$

That is,

$$\text{symmetry is the mathematical expression of indistinguishability}$$

can be read. The connection between gauge symmetry and observation and relations is discussed by Rovelli [15].

There is no need to impose symmetry on nature afterwards.

If exchanging or rotating axes that the observer cannot name leaves the observational results unchanged, that is directly a symmetry.

10. The huge symmetry when only t can be read

When only t can be read, only

$$\rho^2 = x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2$$

can be observed.

Considering only the real squared norm, rotating the seven hidden directions leaves ρ unchanged.

Therefore a large symmetry of type

$$O(7)$$

appears.

What matters is that

the fewer the observable dimensions,
the larger the indistinguishability on the hidden side,
and the larger the apparent symmetry

(Fig. 2).

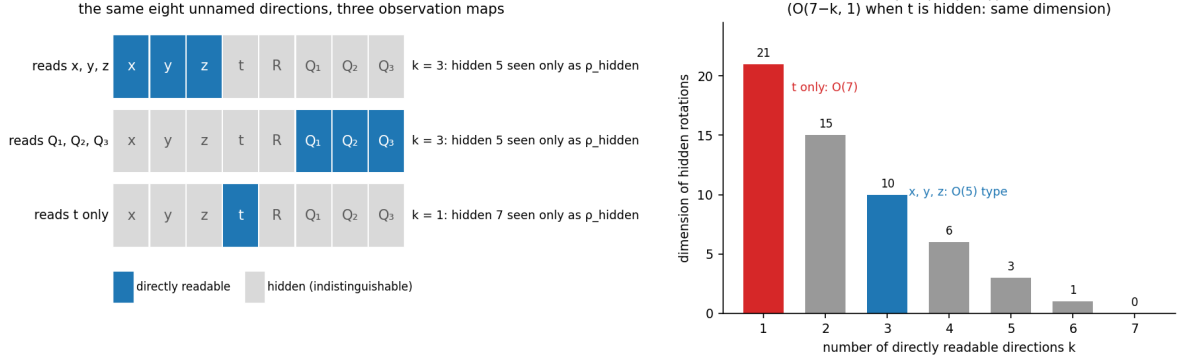


Figure 2. Three observation maps of the same eight directions and the size of the rotational symmetry of the hidden directions. Left: the directions that can be read directly (blue) and the directions that cannot be distinguished (gray) for an observer reading xyz , an observer reading $Q_1Q_2Q_3$ and an observer reading only t . Right: the number k of directly readable directions and the dimension $(8-k)(7-k)/2$ of the rotation group of the hidden directions. Reading only t gives $O(7)$ (21 dimensions); reading xyz gives 10 dimensions. When t is hidden the group becomes $O(7-k, 1)$, but the dimension is the same.

Conversely, when the observational resolution rises and directions that were indistinguishable become distinguishable, the symmetry becomes smaller.

Therefore the interpretation

symmetry breaking = directions that could not be distinguished
becoming distinguishable

also becomes possible.

11. The 3+2 split

Suppose we are observers who directly read

$$(x, y, z).$$

Then on the hidden side the five directions

$$(t, R, Q_1, Q_2, Q_3)$$

remain.

Suppose that observation makes these five directions distinguishable only as the two groups

$$(Q_1, Q_2, Q_3)$$

and

$$(t, R),$$

that is, as

$$\boxed{3+2}$$

If the three internal components, as complex states, are indistinguishable under change of basis, a symmetry of type

$$U(3)$$

becomes a candidate.

Likewise, for the two components a symmetry of type

$$U(2)$$

becomes a candidate.

Therefore the structure

$$U(3) \times U(2)$$

can appear naturally.

Further, imposing a common overall phase / determinant condition and considering

$$S(U(3) \times U(2)),$$

we have

$$S(U(3) \times U(2)) \simeq \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}$$

[10].

This coincides with the global form $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ usually adopted as the gauge group of the Standard Model.

This 3+2 split is the same structure as the one in which $S(U(3) \times U(2)) \subset SU(5)$ remains in the SU(5) grand unified theory [9, 10] (Fig. 3).

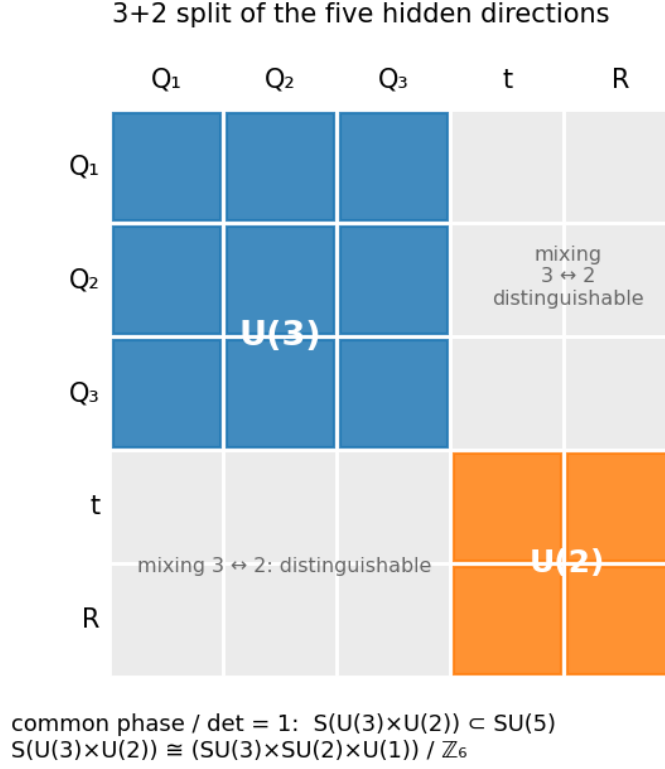


Figure 3. The 3+2 split of the five hidden directions. Mixing within (Q_1, Q_2, Q_3) (blue) corresponds to $U(3)$, mixing within (t, R) (orange) to $U(2)$, and mixing between the two groups (gray) can be distinguished by observation. Imposing a common overall phase / determinant condition gives $S(U(3) \times U(2)) \subset SU(5)$, $S(U(3) \times U(2)) \simeq (SU(3) \times SU(2) \times U(1)) / \mathbb{Z}_6$ [9, 10].

However, at this stage this is **not a derivation**.

In particular, the quadratic closure taken as basic in this research,

$$\sum z_n^2 = 0,$$

and the Hermitian norm preserved by the ordinary unitary groups,

$$\sum |z_n|^2,$$

are not the same.

Obtaining $U(n)$ from eight real directions requires newly assuming a complex structure, and this paper does not derive it.

Therefore, to claim

$$3 + 2 \rightarrow SU(3) \times SU(2) \times U(1),$$

it will be necessary to make clear which observable is taken as the invariant.

Still, it is noteworthy that from the observational structure

3 indistinguishable directions + 2 indistinguishable directions

alone, a mathematical structure extremely close to the Standard Model gauge group appears as a candidate.

12. Reversing the viewpoint

In the usual description,

space, time, mass, charge, color charge

are introduced from the start as different kinds of physical quantities.

Then symmetry is required to connect them.

In this thought experiment the order is reversed.

What exists first is only the unnamed state space

$$(u_1, \dots, u_8)$$

The observer chooses from it a subspace it can distinguish directly.

The rest cannot be distinguished directly.

And the results of projecting them onto the observable space are named

time, mass, charge, color charge.

Then

a physical quantity is not an attribute proper to an axis
but a reading name that depends on the observation map

Likewise,

symmetry is not a constraint added to nature
but the existence of degrees of freedom the observer cannot distinguish

13. Back to the fine-structure constant

Here we return to the fine-structure constant, our starting point.

In the hydrogen ground state,

$$\alpha = \frac{v}{c}.$$

At first this looked like a mere velocity ratio.

But in the view of this thought experiment,

$$v$$

and

c

may also be ratios of reading an unnamed higher-dimensional gradient from different observation directions.

Therefore the possibility arises that it can be interpreted as

$\alpha = \text{ratio of gradient components onto two observation directions}$

Then the fine-structure constant may not be merely “the strength of the electromagnetic interaction” but

$\text{an angle or ratio arising when a higher-dimensional state is projected onto a particular observed subspace}$

At this stage it is undetermined which of

$Q_i/t, \quad Q_i/R, \quad |\mathbf{Q}|/R, \quad \text{or another projection angle}$

it corresponds to.

But at least, starting from the historical fact

$$\alpha = \frac{v}{c},$$

we place a starting point for looking for α as a purely geometric ratio.

14. Verification items from this thought experiment

At least the following verifiable directions arise.

First,

$\text{from the same eight-dimensional dynamics, different effective physics should be obtained only by changing the observed subspace}$

Therefore, without changing the interaction law at all, the effective motion can be computed numerically for each case, such as reading

$xyz,$

reading

$Q_1 Q_2 Q_3,$

or reading only

$t.$

Second,

the fewer the observable dimensions,
the larger the effective symmetry should be

Third,

when the observational resolution rises and
previously indistinguishable internal directions are separated,
the symmetry should break

Fourth, at least some of mass, charge and color charge may be reproducible as

not different basic attributes but
different projected components of the same unnamed state space

Fifth, it can be tested whether the Standard Model symmetry of type

$$SU(3) \times SU(2) \times U(1)$$

can arise from

the observational indistinguishability classes
of 3 directions and 2 directions

15. Conclusion

This thought experiment started by questioning the meaning of one dimensionless constant,

$$\alpha \simeq 1/137$$

In the hydrogen ground state,

$$\alpha = \frac{v}{c}.$$

Reading this not from the Lorentz factor but from

$$r^2 + \tau^2 + (it)^2 = 0,$$

the light-cone equation with one axis added, α appears as a simple direction ratio.

Extending this to

$$x^2 + y^2 + z^2 + R^2 + Q_1^2 + Q_2^2 + Q_3^2 = t^2$$

gives the eight-dimensional structure

$$(x, y, z, t, R, Q_1, Q_2, Q_3).$$

But there is no need to give these eight axes physical names from the start.

Essentially there exists only the unnamed state space

$$(u_1, \dots, u_8),$$

and it can be thought that physical meanings such as

space, time, mass, charge, color

arise according to

which axes are read as the observable space

Then

different physics = different basic laws

need not hold.

Rather,

different physics = different observation maps of the same unnamed physics

may hold.

Further, if we read

symmetry = transformations the observer cannot distinguish

the reason why standard theory requires strong gauge symmetries can also be understood in another form.

And symmetry breaking may be

directions that could not be distinguished
becoming distinguishable

In this view, time, mass, charge, color charge and gauge symmetry are not independent concepts from the start.

They may all be grasped in a unified way as

the results of reading one unnamed higher-dimensional state
from a limited observed subspace

The last question

If the physical law on the eight-dimensional side were only one, closed by

$$\sum_{a=1}^8 z_a^2 = 0$$

alone, then is the very fact that we call

xyz

“space”,

t

“time”,

R

“mass and curvature”, and

Q_1, Q_2, Q_3

“charge and color charge”

a physical law, or merely the observer’s choice?

Reproducing the figures

Figures 1–3 are schematics without data. Running `figures/make_figures.py` (Python 3.9, `numpy`, `matplotlib`) generates SVG and PNG files with English labels in the same folder.

Fig.	File
1	<code>figures/fig01_lightcone_central_projection.{svg,png}</code>
2	<code>figures/fig02_observers_symmetry.{svg,png}</code>
3	<code>figures/fig03_split_3_2.{svg,png}</code>

Record of AI involvement

- **ChatGPT** (2026-10-08): first draft of this paper (structure and text of §1–§15 and “The last question”).
 - **Claude** (2026-10-08): correction of escapes produced by export from Google Docs, rewriting §3 to start from the light-cone equation, unifying the signs of §4, §6 and §15 to a single imaginary axis t , the correspondence between the mass shell and Kaluza-Klein in §5, the statements on the global form, $SU(5)$ and complex structure in §11, addition of the abstract, §0 and references with bibliographic checks (引用文献台帳_20261008.md), the correspondence with central projection in §4, Figures 1–3 and the script `figures/make_figures.py`, and this English translation. Design and judgment are the author’s.
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This research series

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